Notes: Banerjee, Dasgupta, and Kim (JF, 2008)

Buyer–Supplier Relationships and the Stakeholder Theory of Capital Structure

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1 Big picture

- **Research question.** Do buyer–supplier relationships affect corporate capital structure? More specifically, do firms use lower leverage when their customers or suppliers make relationship-specific investments that would be damaged by financial distress or liquidation?
- **Main idea.** The stakeholder theory of capital structure says that leverage affects not only shareholders and debtholders, but also nonfinancial stakeholders. Customers, suppliers, workers, and other partners care about whether the firm will survive, because liquidation can destroy relationship-specific investments.
- **Empirical setting.** The paper studies U.S. firms from 1979 to 1997 and uses Compustat's principal-customer disclosures. A principal customer is a customer that accounts for at least 10 percent of a supplier's sales or is otherwise important to the supplier's business.
- **Customer-side prediction.** A customer firm should maintain lower leverage when it buys more from dependent suppliers, especially when both the customer and suppliers are in durable-goods industries where customized products and specific investments are more important.
- **Supplier-side prediction.** A supplier firm should maintain lower leverage when a large share of its sales goes to principal customers, because losing those customers can destabilize the supplier.
- **Government-customer prediction.** Sales to government customers should matter less for supplier leverage because government buyers are less likely to default and may be more stable sources of demand.
- **Main finding.** The evidence supports the stakeholder theory mainly in durable-goods industries. Durable-goods customers that rely more on dependent durable-goods suppliers have lower leverage, and durable-goods suppliers that rely more on nongovernment principal customers also have lower leverage.
- **Economic takeaway.** Capital structure is partly a commitment device in product-market relationships. Firms keep debt low when doing so helps sustain relationship-specific investment and protects value in bilateral supply chains.

2 Incremental contribution of the paper

- **Direct buyer–supplier measurement.** Earlier evidence, especially Titman and Wessels (1988), used durable-goods industry membership as a broad proxy for product uniqueness. This paper uses actual principal-customer relationships to measure bilateral dependence at the firm level.
- **Two-sided supply-chain test.** The paper studies both sides of the relationship: how dependent suppliers affect customers' leverage and how principal customers affect suppliers' leverage.
- **Within-industry variation.** The empirical design exploits variation inside durable-goods industries, rather than simply comparing durable and nondurable industries.
- **Stakeholder theory in large samples.** The paper provides large-sample evidence that nonfinancial stakeholders shape financing choices, not just anecdotal or case-based evidence.

- **Durable-sector mechanism.** The results are strongest exactly where the theory predicts: durable-goods sectors, where customized products, after-sales support, and asset specificity are more important.
- **Government-customer placebo-like contrast.** The paper separates nongovernment and government principal customers. Nongovernment customers matter for supplier leverage; government customers generally do not. This helps distinguish dependence risk from mere sales concentration.
- **Zero-leverage insight.** The paper links principal-customer dependence to the probability that suppliers hold zero long-term debt, contributing to the literature on why some firms have no leverage.
- **Dynamic evidence.** The paper studies past relationship histories, five-year leverage changes, dynamic panel specifications, and customer-relationship terminations, making the evidence more persuasive than a simple cross-section.

3 Data and measurement

3.1 Sample and principal-customer data

The sample consists of U.S. manufacturing firms in Compustat from 1979 to 1997. The paper stops in 1997 because SFAS 131 changed segment disclosure rules after that period.

- **Principal-customer source.** The paper uses the Business Information File of Compustat 98.
- **Disclosure rule.** Under SFAS No. 14, firms disclosed the names of principal customers if sales to a customer exceeded 10 percent of total revenue or if the firm considered the customer important to its business.
- **Supplier data.** A firm reporting principal customers is treated as a supplier firm.
- **Customer data.** By inverting the supplier reports, the authors identify Compustat-listed firms that are principal customers of Compustat-listed manufacturing suppliers.
- **Final relationship sample.** The paper reports 22,785 manufacturing firm-years with at least one large customer and 35,391 large-customer observations.
- **Customer types.** Large customers are classified as Compustat public companies, private companies, government customers, or unidentified customers.

Interpretation. The data allow the authors to observe which firms depend on which customers, rather than inferring stakeholder dependence only from industry-level characteristics.

3.2 Durable versus nondurable sectors

The paper follows Titman and Wessels in defining durable-goods manufacturing firms as firms with primary SIC codes from 3400 to 3990. Nondurable-goods manufacturing firms have primary SIC codes from 2000 to 3399.

- **Durable goods.** Products are more likely to be unique, customized, long-lived, and supported by relationship-specific assets.
- **Nondurable goods.** Products are more likely to be standardized or easier to replace.
- **Nonmanufacturing customers.** Some principal customers are outside manufacturing; these are analyzed separately on the customer side.

Interpretation. The durable/nondurable split is central to identification. The theory predicts stakeholder leverage effects mainly where relationship-specific investment is important.

3.3 Customer-side dependence variables

The main customer-side variables measure how much of a customer's input purchases come from dependent suppliers.

- **Purchases/COGS – all identified suppliers.** Total purchases from all Compustat-listed manufacturing suppliers that identify the firm as a principal customer, divided by the customer's cost of goods sold.
- **Purchases/COGS – durable suppliers.** The same ratio, using only durable-goods manufacturing suppliers.
- **Purchases/COGS – nondurable suppliers.** The same ratio, using only nondurable-goods manufacturing suppliers.

Interpretation. These variables are meant to capture the extent to which a customer relies on suppliers for whom the customer is economically important. A high value means the customer has a network of stakeholders that may need assurance about the customer's financial health.

3.4 Supplier-side dependence variables

The main supplier-side variables measure the share of the supplier's sales going to principal customers.

- **BIGALL.** Proportion of sales to all principal customers.
- **BIGNONGOV.** Proportion of sales to nongovernment principal customers, including Compustat firms, private companies, and unidentified customers.
- **BIGGOVT.** Proportion of sales to government principal customers.

Interpretation. A high value of BIGALL means that the supplier is less diversified and more exposed to losing a key relationship. The split between BIGNONGOV and BIGGOVT tests whether the leverage effect is driven by demand-risk concerns.

3.5 Capital structure variables and controls

The main dependent variable is long-term market leverage:

$$LDM = \frac{\text{Total long-term debt}}{\text{Total debt} + \text{Market value of equity}}.$$

The paper also checks robustness using book long-term leverage:

$$LDB = \frac{\text{Total long-term debt}}{\text{Book value of total assets}}.$$

The control variables include Tobin's Q, log sales, fixed assets over total assets, return on assets, R&D over sales, SG&A over sales, a dividend dummy, and sales over assets.

Interpretation. The controls are standard capital-structure determinants, while sales/assets is especially important here because it proxies for outsourcing or low vertical integration.

4 Theory and hypotheses

4.1 Stakeholder theory of capital structure

The stakeholder theory starts from the idea that liquidation imposes costs on parties other than investors. Customers may lose support, spare parts, warranties, or customized supply. Suppliers may lose orders and find it difficult to redeploy dedicated assets. These costs can feed back into firm value because stakeholders may refuse to invest in the relationship unless the firm is financially stable.

- High leverage raises the probability of financial distress or liquidation.
- Financial distress can destroy relation-specific investments.
- Stakeholders anticipate this risk and adjust their willingness to invest, supply, buy, or cooperate.
- A lower debt ratio can therefore be valuable because it reassures stakeholders.

Economic intuition. In a simple trade-off model, leverage is chosen by balancing tax benefits and expected distress costs. This paper broadens distress costs to include lost stakeholder investment in buyer–supplier relationships.

4.2 Hypothesis 1: customer firms

Hypothesis 1. A customer firm's debt ratio should be decreasing in the importance of its purchases from dependent suppliers. This effect should be stronger for customers in durable-goods industries and for purchases from dependent suppliers in durable-goods industries.

The logic is:

- Dependent suppliers may need to make customer-specific investments.
- Those investments are risky if the customer is financially fragile.
- Customers can encourage supplier commitment by choosing lower leverage.
- The effect should be strongest when the inputs are customized durable goods.

4.3 Hypothesis 2: supplier firms

Hypothesis 2. A supplier firm's debt ratio should be decreasing in the importance of its sales to principal customers. This effect should be stronger for suppliers in durable-goods industries.

The logic is:

- A supplier that sells a large share of output to a few customers is not diversified.
- Losing a principal customer can cause a sharp demand shock.
- Durable-goods suppliers may have assets that are hard to redeploy.
- Low leverage gives the supplier financial slack against relationship breakdown.

4.4 Hypothesis 3: government customers

Hypothesis 3. Supplier leverage should be less affected by sales to government principal customers than by sales to nongovernment principal customers.

The logic is:

- Government buyers are less likely to default.
- Government demand may be more stable.
- Government customers may be less likely to terminate relationships in ways that trigger supplier distress.
- Government procurement may focus more on price and stated product requirements than on supplier leverage.

Interpretation. If government-customer sales do not reduce supplier leverage, then the main result is less likely to reflect generic customer concentration and more likely to reflect distress risk from private-sector relationships.

5 Methodology and empirical specifications

5.1 Matched comparisons

The paper first compares customer and supplier firms with size- and industry-matched control firms. This addresses the fact that customer firms are generally much larger and older, while supplier firms are generally smaller and younger.

- Supplier firms are matched to control firms in the same two-digit SIC industry and similar size range.
- Customer firms are matched to similar firms in the same industry and size range.
- The matched comparisons show that both customer and supplier firms have lower debt ratios than comparable firms.

Interpretation. These comparisons motivate the regression analysis but do not prove causality, because relationship firms differ in many other characteristics.

5.2 Partial-adjustment leverage model

The main regressions use a dynamic partial-adjustment framework. Let d_{it} be the debt ratio and d^*_{it} the target debt ratio:

$$d_{it} - d_{it-1} = \beta(d^*_{it} - d_{it-1}),$$

where

$$d^*_{it} = \sum_k \alpha_k X_{kit}.$$

Substitution gives the estimating equation:

$$d_{it} = \beta \sum_k \alpha_k X_{kit} + (1-\beta) d_{it-1}.$$

The model includes lagged leverage, the stakeholder variables, firm controls, firm fixed effects, and year fixed effects. Standard errors are heteroskedasticity-consistent and clustered at the firm level.

Interpretation. The lagged dependent variable captures slow adjustment of capital structure. Firm fixed effects focus the analysis on within-firm changes in buyer–supplier relationships over time.

5.3 Customer-firm regressions

The customer-side regressions relate customer leverage to Purchases/COGS from dependent suppliers. The key prediction is a negative coefficient, especially for durable-goods customers buying from durable-goods suppliers.

The paper estimates separate regressions for:

- durable-goods manufacturing customers;
- nondurable-goods manufacturing customers;
- nonmanufacturing customers;
- smaller versus larger durable-goods customers.

Main result. The negative effect appears mainly for durable-goods customers buying from durable-goods dependent suppliers, and it is strongest for smaller customers.

5.4 Supplier-firm regressions

The supplier-side regressions relate supplier leverage to BIGALL, BIGNONGOV, and BIGGOVT. The key prediction is that sales to principal customers reduce leverage, especially in durable-goods industries and especially for nongovernment principal customers.

Main result. BIGALL and BIGNONGOV are negative and significant for durable-goods suppliers. BIGGOVT is not significantly negative. The same patterns are weak or absent in nondurable-goods suppliers.

5.5 Dynamic and robustness tests

The paper runs several additional tests:

- a no-previous-purchases dummy for whether a customer had no dependent suppliers during the past five years;
- five-year leverage-change regressions that separate annual changes in relationship variables;
- logit regressions for whether suppliers have zero long-term debt;
- Arellano–Bond dynamic panel estimates;
- Davidson–MacKinnon endogeneity checks;
- univariate comparisons around the termination of principal-customer relationships;
- robustness to alternative industry definitions, customer-censoring concerns, and sales-volatility controls.

Interpretation. The robustness tests are designed to show that the results are not driven only by static firm differences, adjustment costs, omitted industry composition, or generic sales volatility.

6 Identification and interpretation

6.1 What the paper can identify

The strongest evidence is that leverage changes systematically with the importance of buyer–supplier relationships, after controlling for firm fixed effects, year effects, and standard capital-structure determinants. The paper identifies an economically meaningful association consistent with stakeholder theory.

Causal interpretation. The design is not a randomized experiment, so the evidence should be read as strong panel evidence rather than definitive causal proof. The most persuasive part is that the effects appear exactly where the theory predicts: durable-goods relationships and nongovernment principal customers.

6.2 Why durable goods matter

Durable-goods firms are likely to produce more unique and customized products. Relationship-specific investments are therefore more likely in durable-goods supply chains.

- Durable customers often buy specialized durable inputs from durable suppliers.
- Nondurable customers more often buy standardized inputs.
- The leverage effects are concentrated in the durable sector.

Interpretation. The durable-sector concentration helps distinguish the stakeholder channel from generic size, profitability, or customer-concentration effects.

6.3 Within-firm changes and slow adjustment

The partial-adjustment model and firm fixed effects ask whether leverage changes as a firm's customer–supplier relationships change over time.

- The no-previous-purchases dummy shows that customers without recent dependent suppliers have higher leverage.
- Five-year change regressions show that leverage responds more to earlier changes in relationship variables than to only contemporaneous changes.
- These patterns fit slow capital-structure adjustment.

Interpretation. Firms do not instantly rebalance leverage, but relationship changes appear to affect leverage targets over several years.

6.4 Economic magnitude

The magnitudes are meaningful relative to the low average debt ratios in the sample.

- In the customer regressions, a 10 percentage-point increase in purchases from durable dependent suppliers is associated with about a 0.5 percentage-point lower debt ratio.
- In the supplier regressions, a firm selling 75 percent of sales to nongovernment principal customers has a long-term debt ratio roughly 1.2 percentage points lower than a supplier without principal customers.
- These effects are large relative to mean and median long-term market debt ratios of roughly 11 percent and 6 percent in several relevant groups.

6.5 Limitations

- Principal-customer disclosures are available only before the SFAS 131 change.
- Compustat captures only public firms reliably, so some private buyer–supplier links may be missed.
- Relationship variables may still be correlated with unobserved changes in firm strategy.
- Durable-goods status is an imperfect proxy for specific investment.
- The paper studies U.S. public-firm data, so the results may not generalize to all firms or all countries.

Bottom line. The paper provides strong evidence that stakeholder relationships are a real determinant of leverage, but the estimates should be interpreted as disciplined reduced-form evidence rather than a fully structural estimate of optimal leverage.

7 Tables and figures

7.1 Tables I and II: Summary statistics and matched comparisons

Read Table I:

- Customer firms are much larger and older than the average manufacturing firm.
- Supplier firms are smaller and younger than customer firms.
- Durable-sector firms generally have lower debt ratios than nondurable-sector firms.
- Customer firms have higher profitability and are more likely to pay dividends.

Read Table II:

- Dependent suppliers have lower market and book leverage than size- and industry-matched controls.
- Customer firms also have lower leverage than their matched controls.
- The differences in leverage remain even after matching on size and industry, motivating multivariate regressions.

Economic message: Firms involved in principal-customer relationships look different from comparable firms, especially in leverage. This is the first descriptive evidence that stakeholder relationships and conservative capital structures are connected.

Table I
Summary Statistics

The mean ($\bar{x}$) and the median (Me) values of firm characteristics for firm-years between 1979 and 1997 in the manufacturing sector in Compustat are reported in Panel A. Panels B and C report the same statistics for the sample of customer and supplier firms, respectively. The variables are described in Appendix B. All variables are censored at the 1st and 99th percentile values.

	(A) All Manufacturing Firms in Compustat				(B) All Customer Firms in Compustat in the Manufacturing Sector				(C) All Supplier Firms in the Manufacturing Sector with Principal Customers			
	Durable Sector		Nondurable Sector		Durable Sector		Nondurable Sector		Durable Sector		Nondurable Sector	
	$\bar{x}$	*Me*	$\bar{x}$	*Me*	$\bar{x}$	*Me*	$\bar{x}$	*Me*	$\bar{x}$	*Me*	$\bar{x}$	*Me*
Total Assets	647	33	1,182	90	5,314	1,515	5,851	2,593	303	23	261	40
Q	2.02	1.39	1.93	1.30	1.77	1.37	2.12	1.63	2.05	1.44	2.19	1.40
FA/TA	0.23	0.21	0.34	0.32	0.26	0.24	0.36	0.36	0.22	0.20	0.30	0.29
ROA	−0.07	0.03	−0.05	0.04	0.04	0.05	0.05	0.07	−0.06	0.03	−0.08	0.03
Margin	−0.49	0.34	−1.86	0.28	0.35	0.33	0.14	0.38	0.20	0.33	−0.78	0.25
AGE	10.87	7.00	12.97	8.00	21.40	20.00	25.97	31.00	11.12	8.00	11.37	7.00
R&D	0.08	0.02	0.08	0.00	0.07	0.05	0.07	0.02	0.09	0.04	0.14	0.00
SGA	0.29	0.24	0.22	0.15	0.25	0.24	0.29	0.24	0.32	0.25	0.23	0.16
Dividend	0.37	0.00	0.52	1.00	0.60	1.00	0.82	1.00	0.22	0.00	0.30	0.00
CV(Sales)	1.05	1.02	1.05	1.01	1.04	1.02	1.02	1.01	1.05	1.02	1.06	1.02
S/A	1.92	1.31	2.01	1.37	1.40	1.20	1.33	1.13	1.29	1.23	1.34	1.28
LDM	0.11	0.06	0.15	0.11	0.11	0.09	0.11	0.09	0.10	0.05	0.13	0.08
LDB	0.15	0.10	0.20	0.16	0.14	0.12	0.17	0.14	0.15	0.08	0.19	0.14
Number of Firms	3,202		4,114		416		292		2,430		1,445	
Number of Obs[a]	34,716		27,528		1,840		1,170		14,474		7,643	

[a]The figures are the numbers of firm-years for which data on total assets are available. However, there are some missing values for most of the other variables.

Table I: Summary statistics for manufacturing, customer, and supplier firms.

Table II
Comparison of Firm Characteristics of Dependent Suppliers and Customers with their Respective Control Group Firms

The mean ($\bar{x}$) and the median (Me) values of firm characteristics for the sample of dependent supplier firms in the manufacturing sector and the corresponding control group are shown in Panel A and those for the sample of customer firms and its control group are shown in Panel B. The null hypotheses of no difference between mean firm characteristics of the samples and their respective control groups are tested and the t-values are shown. The control groups have the same number of observations and same industry composition as their respective supplier or customer sample. Their median firm sizes, measured by deflated total assets, are also very similar. Refer to Section II of the paper for a detailed explanation of how the two control samples are constructed. Variable definitions are in Appendix B. All variables are Winsorized at the 0.5 percentile on both tails. *** (**, *) denotes significance at the 1% (5%, 10%) level.

	(A) Comparison of Dependent Supplier Characteristics					(B) Comparison of Customer Firm Characteristics				
	Dependent Suppliers		Control Group		Supplier Minus Control = 0	Big Customers		Control Group		Customer Minus Control = 0
	$\bar{x}$	*Me*	$\bar{x}$	*Me*	t-ratio	$\bar{x}$	*Me*	$\bar{x}$	*Me*	t-ratio
LDM	0.112	0.057	0.122	0.070	−9.14***	0.109	0.085	0.144	0.117	−16.67***
LDB	0.161	0.100	0.167	0.112	−4.05***	0.151	0.128	0.186	0.158	−13.60***
Age	10.54	7.000	11.40	8.000	−12.33***	24.26	28.00	19.84	18.00	15.47***
R&D	0.113	0.019	0.104	0.013	5.535***	0.062	0.038	0.053	0.018	4.252***
SGA	0.286	0.220	0.322	0.254	−20.42	0.241	0.221	0.214	0.183	9.150***
Advert	0.018	0.002	0.024	0.010	−14.26***	0.029	0.014	0.021	0.009	7.161***
ROA	−0.05	0.031	−0.06	0.031	3.89***	0.055	0.063	0.034	0.047	11.37***
Q	2.111	1.430	2.062	1.338	3.14***	1.912	1.475	1.567	1.283	12.54***
FA/TA	0.251	0.218	0.251	0.228	−0.74	0.304	0.286	0.317	0.289	−4.549***
Intang	0.037	0.000	0.040	0.000	−5.48***	0.048	0.000	0.052	0.002	−2.438**
STDE	12.00	3.601	13.80	3.456	−8.46***	459.2	163.6	277.7	83.59	11.39***
CV(Sales)	1.057	1.022	1.058	1.019	−1.18	1.033	1.013	1.029	1.011	3.231***
Dividend	0.231	0.000	0.314	0.000	−26.88***	0.682	1.000	0.699	1.000	−1.837*
S/A	1.275	1.234	1.277	1.253	−0.43	1.233	1.171	1.137	1.087	10.49***
Sales	140.9	30.19	168.1	36.13	−10.27***	6,126	2,378	4,910	1,498	5.537***
Obs			18,830					2,554		

Table II: Dependent suppliers and customer firms versus matched control groups.

7.2 Tables IIIA–C and IV: Relationship importance and duration

Read Table IIIA:

- Purchases from dependent suppliers are most important for smaller customers.
- In the smallest size quartile, dependent-supplier purchases are economically meaningful as a share of materials cost.
- The importance of dependent suppliers declines monotonically with customer size.

Read Tables IIIB and IIIC:

- Purchases from dependent suppliers are not simply a high-sales-growth phenomenon.
- Durable-sector customers mostly purchase from durable dependent suppliers, while nondurable customers mostly purchase from nondurable suppliers.
- This supports the idea that durable customer–durable supplier pairs are where relationship-specific investment is most likely.

Read Table IV:

- Customer–supplier relationships last several years and are clustered in time.
- Durable-goods customers report dependent suppliers for an average of 4.39 years.
- Durable-goods suppliers report principal customers for an average of 5.29 years.

Economic message: The relationships are not one-off transactions. They are persistent enough for capital structure to plausibly respond, but temporary enough for the risk of relationship termination to matter.

Table IIIA
Proportion of Purchases from Dependent Suppliers in the Customer Firm's Materials Cost

Summary statistics for the amount of purchase by customer firms from their dependent suppliers as a proportion of the customer firms' materials cost are reported. Materials cost of a firm is obtained by multiplying the firm's Cost of Goods Sold with the relevant industry's ratio of materials cost to Cost of Goods Sold. Industry data are available at the four-digit SIC level in the National Bureau of Economic Research site. The sample consists of customer firm-years between 1979 and 1997 only and the observations are partitioned into size quartiles.

Size Groups of Customer Firms	All Manufacturing Sector Customer Firms		Durable Goods Sector Customer Firms		Nondurable Goods Sector Customer Firms	
	Mean	Median	Mean	Median	Mean	Median
0–25 percentile	13.0%	7.0%	14.1%	7.0%	10.7%	6%
25–50 percentile	4.0%	1.7%	5.1%	2.1%	2.3%	1.4%
50–75 percentile	2.2%	0.7%	2.9%	0.8%	1.3%	0.3%
75–100 percentile	1.9%	0.7%	2.5%	1.0%	1.0%	0.4%

Table IIIA: Purchases from dependent suppliers by customer size quartile.

Table IIIB
Proportion of Purchases from Dependent Suppliers during High and Low Sales Years

Columns 1 and 2 report the mean and median purchases from dependent suppliers as a fraction of total sales when sales growth in that firm-year is in the highest quartile (high sales growth) or in the lowest quartile (low sales growth) for each firm. The sample consists of customer firm-years between 1979 and 1997.

	(A) Durable Goods Manufacturing Customers		(B) Nondurable Goods Manufacturing Customers		(C) Nonmanufacturing Customers	
	(1) Low Sales Growth	(2) High Sales Growth	(1) Low Sales Growth	(2) High Sales Growth	(1) Low Sales Growth	(2) High Sales Growth
Mean	0.043	0.046	0.028	0.038	0.033	0.039
Median	0.012	0.013	0.006	0.005	0.010	0.010
Number of Firms	509	555	332	329	502	517

Table IIIC
Distribution of Purchases from Durable versus Nondurable Dependent Suppliers

The first column reports the number of years in which a customer firm makes purchases from dependent suppliers in the manufacturing sector between 1979 and 1997. The following three columns show the number of cases in which such purchases were made only from durable goods suppliers, nondurable goods suppliers, and both types of dependent suppliers.

	Number of Obs. with Purchases from Dependent Suppliers	Number of Obs. with Purchases from Durable Dependent Suppliers	Number of Obs. with Purchases from Nondurable Dependent Suppliers	Number of Obs. with Purchases from both Types of Dependent Suppliers
Durable Sector Customers	1,829	1,661	375	207
Nondurable Sector Customers	1,144	367	892	115
Nonmanufacturing Customers	1,803	1,262	776	235

Tables IIIB–C: Purchases during high/low sales years and by supplier sector.

Table IV

Descriptive Statistics for the Frequency and Duration of Customer–Supplier Relationships

Panel A reports summary statistics for the number (proportion) of years a firm, identified at least once as a principal customer, purchases from dependent suppliers and Panel B reports summary statistics for the number (proportion) of years a suppler firm reports sales to principal customers. For each panel, the four columns report the summary statistics of the following items for the respective subgroup of customer and supplier firms:

Column (1): Total number of years a customer (supplier) firm reports a dependent supplier (large customer).

Column (2): Number of *consecutive years* a customer (supplier) firm reports a dependent supplier (large customer). If a customer firm has more than one stretch of consecutive years of purchases from dependent suppliers, the stretch with the maximum number of consecutive years is used.

Column (3): Proportion of years a customer (supplier) reports a dependent supplier (large customer) as a fraction of years the firm appears in the sample.

Column (4): Proportion of *consecutive years* a customer (supplier) firm reports a dependent supplier (large customer) as a fraction of years the firm appears in the sample.

Panel A

	(A) Durable Goods Manufacturing Customers				(B) Nondurable Goods Manufacturing Customers				(C) Nonmanufacturing Customers			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Mean	4.39	3.73	0.35	0.31	3.96	3.36	0.31	0.28	4.16	3.53	0.32	0.29
SD	4.48	4.04	0.27	0.26	3.80	3.24	0.25	0.24	4.38	3.88	0.27	0.25
5th percentile	1	1	0.05	0.05	1	1	0.05	0.05	1	1	0.05	0.05
25th percentile	1	1	0.13	0.11	1	1	0.11	0.10	1	1	0.11	0.11
Median	3	2	0.26	0.21	2	2	0.25	0.20	2	2	0.21	0.17
75th percentile	5	4	0.50	0.42	5	4	0.42	0.37	5	4	0.47	0.36
95th percentile	16	14	1	1	13	11	0.92	0.83	14	12	1	1
Number of Firms	431				299				440			

Panel B

	(A) Durable Goods Manufacturing Suppliers				(B) Nondurable Goods Manufacturing Suppliers			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Mean	5.29	4.73	0.55	0.51	5.96	5.37	0.60	0.56
SD	4.06	3.75	0.29	0.29	4.75	4.49	0.29	0.30
5th percentile	1	1	0.11	0.11	1	1	0.125	0.11
25th percentile	2	2	0.31	0.25	2	2	0.33	0.30
Median	4	4	0.5	0.47	4	4	0.6	0.50
75th percentile	7	6	0.8	0.75	8	7	0.875	0.831
95th percentile	14	12	1	1	17	16	1	1
Number of Firms	1,445				2,430			

Table IV: Frequency and duration of customer–supplier relationships.

Table V

Pair-Wise Correlations between the Explanatory Variables for the Sample of All Firms

The sample contains all firm-years for firms between 1979 and 1997 in the manufacturing sector in Compustat that have no missing values for any of the listed variables. There are 59,526 such observations. The list of variables includes all the independent variables used in the subsequent regressions. The definitions of these variables are in Appendices A and B. The significance level is reported below the correlation coefficients.

	Q	Log (Sales)	FA/TA	ROA	Age	R&D	SGA	Dividend Dummy	S/A	BIGALL
Log(Sales)	−0.255									
	0.000									
FA/TA	−0.052	0.299								
	0.000	0.000								
ROA	−0.214	0.462	0.068							
	0.000	0.000	0.000							
R&D	0.172	−0.412	−0.168	−0.529	−0.220					
	0.000	0.000	0.000	0.000	0.000					
SGA	0.133	−0.403	−0.249	−0.420	−0.219	0.283				
	0.000	0.000	0.000	0.000	0.000	0.000				
Dividend Dummy	−0.193	0.560	0.258	0.284	0.462	−0.234	−0.252			
	0.000	0.000	0.000	0.000	0.000	0.000	0.000			
S/A	0.018	0.171	−0.011	0.162	0.062	−0.373	−0.225	0.079		
	0.000	0.000	0.011	0.000	0.000	0.000	0.000	0.000		
BIGALL	0.079	−0.247	−0.111	−0.112	−0.163	0.201	0.015	−0.216	−0.055	
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	
Purchases/ COGS All Suppliers	−0.009	0.074	−0.024	0.021	0.017	0.008	0.000	0.003	0.004	0.006
	0.028	0.000	0.000	0.000	0.000	0.043	0.960	0.488	0.375	0.175

Table V: Pair-wise correlations among explanatory variables.

7.3 Tables VI and VII: Customer leverage and dependent suppliers

Read Tables VI and VII:

- The dependent variable is long-term market leverage.
- In durable-goods manufacturing customers, Purchases/COGS from all identified suppliers is negative.
- The negative effect is driven by purchases from durable-goods dependent suppliers.
- The effect is strongest among smaller durable-goods customers.
- The same relationship is weak or absent for nondurable manufacturing customers and nonmanufacturing customers.
- Table VII shows that the results are similar when the sample is expanded to all firms rather than only firms ever identified as principal customers.

Economic message: Customer firms appear to choose conservative leverage when their dependent suppliers are likely to make specific investments. This is the paper's core customer-side evidence.

Table VI

The Effect of Dependent Suppliers on the Market Leverage of Customer Firms

The following model is estimated: $LDM_{it} = \alpha + \gamma LDM_{it-1} + \Sigma_k \delta_k B_{kit} + \Sigma_l \delta_l F_{lit} + \kappa_i + u_{it}$, where the market leverage, LDM_{it}, is defined as the ratio of long-term debt to the market value of equity plus total debt. B_{kit} represents purchases from dependent manufacturing suppliers from different sectors as a fraction of *Cost of Goods Sold* of the customer firms. Please see Appendix A for exact definitions of these variables. F_{lit} refers to control variables in the regressions. It includes log of total sales (*Log(Sales)*), profitability (*ROA*), tangibility (*FA/TA*), growth opportunities (*Q*), R&D to sales (*R&D*), selling general and administrative expense (*SGA*), dividend dummy (*Dividend*), and total sales to total assets (*S/A*). The definitions of these variables are in Appendix B. The regressions also include firm and year dummies, but their coefficients are not reported. The sample contains all firm-years for firms in Compustat that are identified as a principal customer by at least one other Compustat firm *for at least one year* between 1979 and 1997. The control variables are censored at the 1st and 99th percentile values to eliminate influence of outliers. Panel A shows the results for the manufacturing firms in durable goods industries only (primary SIC codes from 3,400 to 3,990). In column 3 (respectively, column 4), we drop all customer firms whose median size is above median (respectively, below median) among all customer firms. Panel B replicates the analysis for manufacturing firms in the nondurable goods industries only (SIC codes between 2,000 and 3,400) and Panel C shows the result for nonmanufacturing customer firms (all other sectors except the financial sector). The *t*-values, given in parentheses, are based on standard errors that are heteroskedasticity-consistent and allow clustering at the firm level. *** (**,*) denotes significance at the 1% (5%, 10%) level.

	(A) Durable Goods Manufacturing Customers				(B) Nondurable Goods Manufacturing Customers		(C) Nonmanufacturing Customers	
	(1) All Firms	(2) All Firms	(3) Smaller Customer Firms	(4) Larger Customer Firms	(1) All Firms	(2) All Firms	(1) All Firms	(2) All Firms
LDM_{t-1}	0.518	0.517	0.497	0.527	0.536	0.536	0.545	0.545
	(19.73)***	(19.72)***	(13.93)***	(13.17)***	(17.80)***	(17.81)***	(18.65)***	(18.63)***
Purchases/COGS	−0.040				−0.012		0.089	
All identified suppliers	(−1.54)				(−0.30)		(1.44)	
Purchases/COGS		−0.048	−0.071	0.018		0.047		0.110
Durable suppliers		(−1.81)*	(−2.43)**	(−0.26)		(0.92)		(1.44)
Purchases/COGS		0.120	0.160	−0.414		−0.039		0.015

Table VI, part 1: Dependent suppliers and customer leverage among principal-customer firms.

Q	−0.013	−0.013	−0.011	−0.019	−0.010	−0.010	−0.022	−0.022
	(−6.78)***	(−6.77)***	(−5.31)***	(−4.22)***	(−4.91)***	(−4.92)***	(−7.04)***	(−7.03)***
Log(Sales)	0.016	0.016	0.017	0.014	0.018	0.018	0.011	0.012
	(3.72)***	(3.76)***	(2.93)***	(3.15)***	(4.53)***	(4.56)***	(2.10)**	(2.12)**
FA/TA	0.088	0.089	0.157	−0.006	−0.012	−0.012	0.051	0.051
	(2.14)**	(2.15)**	(2.54)**	(−0.16)	(−0.44)	(−0.45)	(1.90)*	(1.90)*
ROA	−0.051	−0.051	−0.061	−0.071	−0.032	−0.032	−0.080	−0.080
	(−2.82)***	(−2.83)***	(−3.06)***	(−1.51)	(−1.06)	(−1.06)	(−3.08)***	(−3.12)***
R&D	0.011	0.011	−0.031	0.079	−0.015	−0.015	−0.069	−0.070
	(0.29)	(0.28)	(−0.76)	(0.83)	(−0.61)	(−0.61)	(−1.03)	(−1.04)
SGA	−0.035	−0.035	−0.064	−0.007	0.010	0.010	−0.012	−0.012
	(−1.38)	(−1.38)	(−1.80)*	(−0.24)	(0.79)	(0.81)	(−0.51)	(−0.49)
Dividend	−0.015	−0.015	−0.012	−0.020	−0.008	−0.008	−0.018	−0.018
	(−2.13)**	(−2.15)**	(−1.28)	(−1.84)*	(−0.77)	(−0.78)	(−1.90)*	(−1.91)*
S/A	−0.041	−0.041	−0.044	−0.035	−0.056	−0.055	−0.033	−0.033
	(−4.61)***	(−4.61)***	(−4.82)***	(−2.43)**	(−5.64)***	(−5.59)***	(−4.93)***	(−4.94)***
Constant	0.008	0.006	0.022	0.019	0.010	0.008	0.072	0.072
	(0.25)	(0.22)	(0.64)	(0.48)	(0.28)	(0.25)	(1.57)	(1.56)
Firms	408	408	230	178	266	266	394	394
Observations	4,675	4,675	2,309	2,366	3,414	3,414	4,436	4,436
R-squared	0.72	0.72	0.70	0.76	0.78	0.78	0.80	0.80

Table VI, part 2: Continued customer-leverage regressions.

Table VII

The Effect of Dependent Suppliers on Market Leverage of Customer Firms (Sample of All Firms)

The following model is estimated: $LDM_{it} = \alpha + \gamma LDM_{it-1} + \Sigma_k \delta_k B_{kit} + \Sigma_l \delta_l F_{lit} + \kappa_i + u_{it}$, where the market leverage, LDM_{it}, is defined as the ratio of long-term debt to the market value of equity plus total debt. B_{kit} represents purchases from manufacturing suppliers from different sectors as a fraction of *Cost of Goods Sold* of the customer firms. Please see Appendix A for exact definitions of these variables. F_{lit} refers to control variables in the regressions. It includes log of total sales (*Log(Sales)*), profitability (*ROA*), tangibility (*FA/TA*), growth opportunities (*Q*), R&D to sales (*R&D*), selling general and administrative expense (*SGA*), dividend dummy (*Dividend*), and total sales to total assets (*S/A*). The definitions of these variables are in Appendix B. The regressions also include firm and year dummies, but their coefficients are not reported. The sample contains all firms in Compustat between 1979 and 1997. The control variables are censored at the 1st and 99th percentile values to eliminate influence of outliers. Panel A shows the results for the manufacturing firms in durable goods industries only (primary SIC codes from 3,400 to 3,990). In column 3 (respectively, column 4), we drop all customer firms whose median size is above median (respectively, below median) among all customer firms. Panel B replicates the analysis for manufacturing firms in the nondurable goods industries only (SIC codes between 2,000 and 3,400) and Panel C shows the result for nonmanufacturing customer firms (all other sectors except the financial sector). The *t*-values, given in parentheses, are based on standard errors that are heteroskedasticity-consistent and allow clustering at the firm level. *** (**,*) denotes significance at the 1% (5%, 10%) level.

	(A) Durable Goods Manufacturing Customers				(B) Nondurable Goods Manufacturing Customers		(C) Nonmanufacturing Customers	
	(1) All Firms	(2) All Firms	(3) Smaller Customer Firms	(4) Larger Customer Firms	(1) All Firms	(2) All Firms	(1) All Firms	(2) All Firms
LDM_{t-1}	0.472	0.472	0.466	0.469	0.481	0.481	0.453	0.453
	(34.94)***	(34.94)***	(32.94)***	(32.44)***	(31.38)***	(31.38)***	(48.21)***	(48.21)***
Purchases/COGS	−0.046				−0.008		0.073	
All identified suppliers	(−1.85)*				(−0.20)		(1.05)	
Purchases/COGS		−0.056	−0.075	0.025		0.051		0.091
Durable suppliers		(−2.17)**	(−2.70)***	(0.37)		(1.18)		(1.06)

Table VII, part 1: Dependent suppliers and customer leverage in the full Compustat sample.

Q	−0.009	−0.009	−0.009	−0.009	−0.009	−0.009	−0.011	−0.011
	(−14.3)***	(−14.3)***	(−13.7)***	(−13.3)***	(−11.9)***	(−11.9)***	(−17.7)***	(−17.7)***
Log(Sales)	0.015	0.015	0.015	0.015	0.017	0.017	0.015	0.015
	(9.81)***	(9.82)***	(9.29)***	(9.54)***	(9.10)***	(9.11)***	(12.61)***	(12.61)***
FA/TA	0.144	0.144	0.155	0.142	0.112	0.112	0.106	0.106
	(10.51)***	(10.52)***	(10.78)***	(10.43)***	(8.16)***	(8.16)***	(13.92)***	(13.92)***
ROA	−0.040	−0.040	−0.038	−0.038	−0.026	−0.026	−0.016	−0.016
	(−8.06)***	(−8.06)***	(−7.72)***	(−7.59)***	(−4.17)***	(−4.17)***	(−4.11)***	(−4.11)***
R&D	−0.005	−0.005	−0.007	−0.004	0.013	0.013	0.002	0.002
	(−0.54)	(−0.55)	(−0.76)	(−0.41)	(1.45)	(1.45)	(0.27)	(0.27)
SGA	−0.020	−0.020	−0.020	−0.018	−0.006	−0.006	−0.006	−0.006
	(−3.17)***	(−3.17)***	(−3.09)***	(−2.83)***	(−1.07)	(−1.06)	(−1.07)	(−1.07)
Dividend	−0.012	−0.012	−0.012	−0.012	−0.014	−0.014	−0.009	−0.009
	(−3.81)***	(−3.81)***	(−3.52)***	(−3.60)***	(−3.35)***	(−3.35)***	(−3.01)***	(−3.02)***
S/A	−0.029	−0.029	−0.028	−0.028	−0.035	−0.035	−0.025	−0.025
	(−10.8)***	(−10.8)***	(−10.2)***	(−10.1)***	(−9.22)***	(−9.22)***	(−14.3)***	(−14.3)***
Constant	0.049	0.049	0.053	0.048	0.055	0.055	0.039	0.039
	(7.12)***	(7.11)***	(7.75)***	(6.84)***	(5.43)***	(5.42)***	(5.85)***	(5.85)***
Firms	3,357	3,357	3,179	3,127	2,517	2,517	6,734	6,734
Observations	25,221	25,221	22,855	22,912	19,751	19,751	44,388	44,388
R-squared	0.72	0.72	0.72	0.72	0.76	0.76	0.78	0.78

Table VII, part 2: Continued full-sample customer regressions.

7.4 Tables VIII and IX: Supplier leverage and principal customers

Read Tables VIII and IX:

- The dependent variable is long-term market leverage.
- BIGALL is negative and significant for durable-goods suppliers.
- BIGNONGOV is more strongly negative than BIGALL, showing that nongovernment principal customers drive the effect.
- BIGGOVT is not significantly negative, consistent with the prediction that government customers are more stable.
- The nondurable-goods supplier results are weak or insignificant.

Economic message: Suppliers with risky private-sector customer dependence maintain lower leverage, especially when they produce durable or customized goods. Government-customer dependence does not create the same leverage response.

Table VIII

The Effect of Sales to Principal Customers on the Market Leverage of Dependent Suppliers

The following model is estimated: $LDM_{it} = \alpha + \gamma LDM_{it-1} + \Sigma_k \delta_k B_{kit} + \Sigma_l \delta_l F_{lit} + \kappa_i + u_{it}$, where the market leverage, LDM_{it} is defined as the ratio of long-term debt to the market value of equity plus total debt. B_{kit} is the proportion of sales to each category of principal customer as a percentage of total sales. The exact definitions of these variables are shown in Appendix A. F_{lit} refers to the control variables in the regressions. It includes log of total sales (*Log(Sales)*), profitability (*ROA*), tangibility (*FA/TA*), growth opportunities (*Q*), R&D to sales (*R&D*), selling general and administrative expense (*SGA*), dividend dummy (*Dividend*), and total sales to total assets (*S/A*). The definitions of these variables are in Appendix B. The regressions also include firm and year dummies, but their coefficients are not reported. The sample contains all manufacturing firms in Compustat that have been a supplier to a large customer *at least once* between 1979 and 1997. The control variables are censored at the 1st and 99th percentile values to eliminate influence of outliers. Panel A reports the results for manufacturing firms in durable goods industries only (primary SIC codes from 3,400 to 3,990) and Panel B shows the analysis for manufacturing firms in the nondurable goods industries only (SIC codes between 2,000 and 3,400). The *t*-values, given in parentheses, are based on standard errors that are heteroskedasticity-consistent and allow clustering at the firm level. *** (**) denotes significance at the 1% (5%) level.

	(A) Durable Goods Firms		(B) Nondurable Goods Firms	
	(1)	(2)	(1)	(2)
LDM_{t-1}	0.465	0.465	0.473	0.473
	(27.85)***	(27.86)***	(21.23)***	(21.19)***
BIGALL	−0.013		−0.006	
	(−2.44)**		(−1.04)	
BIGNONGOV		−0.016		−0.008
		(−2.88)***		(−1.34)
BIGGOVT		0.019		−0.005
		(1.41)		(−0.14)
Q	−0.010	−0.010	−0.008	−0.008
	(−12.85)***	(−12.86)***	(−8.99)***	(−8.97)***
Log(Sales)	0.011	0.011	0.016	0.016
	(5.53)***	(5.52)***	(6.61)***	(6.54)***
FA/TA	0.167	0.168	0.123	0.123
	(10.39)***	(10.41)***	(6.59)***	(6.60)***
ROA	−0.043	−0.043	−0.021	−0.020
	(−7.00)***	(−6.92)***	(−2.65)***	(−2.63)***
R&D	−0.030	−0.030	0.011	0.011
	(−2.63)***	(−2.67)***	(1.09)	(1.08)
SGA	−0.031	−0.031	−0.009	−0.009
	(−3.79)***	(−3.73)***	(−1.46)	(−1.43)
Dividend	−0.009	−0.009	−0.014	−0.014
	(−2.56)**	(−2.51)**	(−2.61)***	(−2.61)***
S/A	−0.033	−0.033	−0.034	−0.034
	(−9.58)***	(−9.58)***	(−7.67)***	(−7.67)***
Constant	0.075	0.075	0.069	0.069
	(8.62)***	(8.57)***	(6.02)***	(6.01)***
Firms	2,161	2,161	1,268	1,268
Observations	16,682	16,682	9,946	9,946
R-squared	0.72	0.72	0.75	0.75

Table VIII: Sales to principal customers and leverage among supplier firms.

Table IX

The Effect of Sales to Principal Customers on the Market Leverage of Dependent Suppliers (Sample of All Firms)

The following model is estimated: $LDM_{it} = \alpha + \gamma LDM_{it-1} + \Sigma_k \delta_k B_{kit} + \Sigma_l \delta_l F_{lit} + \kappa_i + u_{it}$, where the market leverage, LDM_{it} is defined as the ratio of long-term debt to the market value of equity plus total debt. B_{kit} is the proportion of sales to each category of principal customer as a percentage of total sales. The exact definitions of these variables are shown in Appendix A. F_{lit} refers to the control variables in the regressions. It includes log of total sales (*Log(Sales)*), profitability (*ROA*), tangibility (*FA/TA*), growth opportunities (*Q*), R&D to sales (*R&D*), selling general and administrative expense (*SGA*), dividend dummy (*Dividend*), and total sales to total assets (*S/A*). The definitions of these variables are in Appendix B. The regressions also include firm and year dummies, but their coefficients are not reported. The sample contains all manufacturing firms in Compustat between 1979 and 1997. The control variables are censored at the 1st and 99th percentile values to eliminate influence of outliers. Panel A reports the results for manufacturing firms in durable goods industries only (primary SIC codes from 3,400 to 3,990) and Panel B shows the analysis for manufacturing firms in the nondurable goods industries only (SIC codes between 2,000 and 3,400). The *t*-values, given in parentheses, are based on standard errors that are heteroskedasticity-consistent and allow clustering at the firm level. *** (**,*) denotes significance at the 1% (5%, 10%) level.

	(A) Durable Goods Firms		(B) Nondurable Goods Firms	
	(1)	(2)	(1)	(2)
LDM_{t-1}	0.472	0.472	0.480	0.480
	(35.21)***	(35.22)***	(30.38)***	(30.35)***
BIGALL	−0.012		−0.004	
	(−2.27)**		(−0.62)	
BIGNONGOV		−0.015		−0.006
		(−2.73)***		(−0.93)
BIGGOVT		0.018		−0.004
		(1.36)		(−0.09)
Q	−0.009	−0.009	−0.009	−0.009
	(−13.86)***	(−13.86)***	(−11.66)***	(−11.64)***
Log(Sales)	0.014	0.014	0.017	0.017
	(8.00)***	(7.99)***	(8.71)***	(8.65)***
FA/TA	0.145	0.145	0.120	0.120
	(10.43)***	(10.45)***	(8.51)***	(8.52)***
ROA	−0.046	−0.046	−0.028	−0.028
	(−8.92)***	(−8.85)***	(−4.31)***	(−4.29)***
R&D	−0.022	−0.022	0.006	0.006
	(−2.23)**	(−2.26)**	(−0.73)	(−0.72)
SGA	−0.029	−0.029	−0.011	−0.011
	(−4.13)***	(−4.09)***	(−1.93)*	(−1.91)*
Dividend	−0.012	−0.012	−0.014	−0.014
	(−3.84)***	(−3.80)***	(−3.52)***	(−3.53)***
S/A	−0.035	−0.035	−0.042	−0.042
	(−12.33)***	(−12.34)***	(−11.98)***	(−11.98)***
Constant	0.067	0.067	0.063	0.063
	(8.75)***	(8.73)***	(6.59)***	(6.58)***
Firms	3,348	3,348	2,477	2,477
Observations	25,200	25,200	19,512	19,512
R-squared	0.73	0.73	0.76	0.76

Table IX: Sales to principal customers and leverage in the full manufacturing sample.

7.5 Tables X and XI: Relationship history and five-year changes

Read Table X:

- The no-previous-purchases dummy is positive and significant for durable-goods customers.
- This means customers with no dependent-supplier history during the past five years have higher leverage.
- The effect is strongest for smaller durable-goods customers.
- Even when current purchases are zero, recent history matters, consistent with slow leverage adjustment.

Read Tables XIA and XIB:

- The five-year change regressions split relationship changes into annual lags.
- For durable-goods customers, past changes in Purchases/COGS predict five-year leverage changes.
- For durable-goods suppliers, changes in BIGALL are negative across multiple lags.
- These effects are absent or much weaker outside durable-goods sectors.

Economic message: Capital structure reacts gradually to customer–supplier relationships. This supports the interpretation that firms adjust target leverage as relationship-specific investment exposure changes.

Table X

The Effect of Past History of Purchases from Dependent Suppliers on Market Leverage of Customer Firms

The dependent variable is LDM_{it}, defined as the ratio of long-term debt to market value of equity plus total debt. The *No previous purchases dummy* is our variable of interest and it takes the value one if there are *no purchases* from dependent suppliers in the past 5 years. We control for lagged market leverage (LDM_{it-1}), log of total sales ($Log(Sales)$), profitability (ROA), tangibility (PA/TA), growth opportunities (Q), R&D to sales ($R\&D$), selling general and administrative expense (SGA), dividend dummy ($Dividend$), and total sales to total assets (S/A). The sample contains all firm-years for firms in Compustat that are identified as a principal customer by at least one other Compustat firm *for at least 1 year* between 1979 and 1997 and this dummy variable is computed only for observations that at least have 5 years of past data. Panel A shows the results for the manufacturing firms in durable goods industries only (primary SIC codes from 3,400 to 3,990), Panel B replicates the analysis for manufacturing firms in the nondurable goods industries only (SIC codes between 2,000 and 3,400), and Panel C shows the result for nonmanufacturing customer firms (all other sectors except the financial sector). We control for firm fixed effects and allow for clustering at firm level. The t-values, given in parentheses, are based on heteroskedasticity-consistent standard errors. *** (**,*) denotes significance at the 1% (5%, 10%) level.

	(A) Durable Goods Manufacturing Customers				(B) Nondurable Goods Manufacturing Customers	(C) Non-manufacturing Customers
	(1) All Customer Firms	(2) Larger Customer Firms	(3) Smaller Customer Firms	(4) All Customer Firms: "No Purchases" Years Only	(1) All Firms	(1) All Firms
LDM_{t-1}	0.457	0.497	0.403	0.482	0.462	0.460
	(15.62)***	(13.23)***	(9.65)***	(12.87)***	(10.51)***	(12.66)***
No previous purchases dummy	0.006	0.003	0.010	0.007	0.001	0.005
	(1.99)**	(0.72)	(2.27)**	(1.65)*	(0.47)	(1.41)
Q	−0.016	−0.026	−0.013	−0.019	−0.008	−0.025
	(−8.73)***	(−5.81)***	(−6.52)***	(−7.03)***	(−5.66)***	(−7.51)***
Log(Sales)	0.022	0.025	0.021	0.020	0.021	0.007
FA/TA	0.129	−0.041	0.245	0.152	0.006	0.045
	(3.40)***	(−1.07)	(4.55)***	(2.68)***	(0.21)	(1.51)
ROA	−0.045	−0.034	−0.064	−0.033	−0.020	−0.089
	(−1.90)*	(−0.75)	(−2.43)**	(−1.04)	(−0.83)	(−3.22)***
AGE	−0.171	−0.195	−0.227	−0.180	−0.279	−0.140
	(−3.39)***	(−3.38)***	(−2.87)***	(−2.69)***	(−6.39)***	(−2.68)***
R&D	−0.026	0.060	−0.099	−0.049	−0.009	−0.138
	(−0.43)	(0.64)	(−1.35)	(−0.60)	(−0.30)	(−2.68)***
SGA	−0.037	0.013	−0.088	−0.032	0.005	0.007
	(−1.42)	(0.54)	(−1.87)*	(−0.77)	(0.49)	(0.40)
Dividend	−0.015	−0.010	−0.015	−0.011	0.000	−0.010
	(2.03)**	(0.88)	(1.61)	(−1.14)	(0.01)	(−1.07)
S/A	−0.045	−0.032	−0.055	−0.037	−0.079	−0.032
	(−4.40)***	(−2.59)***	(−4.95)***	(−2.99)***	(−8.19)***	(−5.78)***
Constant	0.024	0.003	0.064	0.028	0.104	0.154
	(1.02)	(0.07)	(1.89)*	(0.84)	(3.62)***	(5.24)***
Firms	383	173	210	326	247	364
Observations	3,255	1,681	1,574	1,980	2,377	3,085
R-squared	0.79	0.76	0.71	0.74	0.79	0.69

Table X, part 1: Past history of purchases from dependent suppliers.

Table X, part 2: Continued past-history regressions.

Table XIA

The Effects of Year-by-Year Changes in Purchases from Dependent Suppliers on Five-Year Changes in Market Leverage of Customer Firms

The following model is estimated: $\Delta LDM_{it} = \alpha + \Sigma_k \delta_k \Delta B_{kit} + \Sigma_l \delta_l \Delta F_{lit} + \gamma(LDM_{it-5} - LDM^*_{it-5}) + \kappa_i + u_{it}$, where the market leverage, LDM_{it}, is defined as the ratio of long-term debt to the market value of equity plus total debt. The symbol Δ stands for 5-year changes in the variable. $LDM_{it-5} - LDM^*_{it-5}$ is the deviation of market leverage from its predicted value at year $(t-5)$. LDM^*_{it} is predicted using the model, $LDM_{it} = \alpha + \Sigma_k \delta_k B_{kit} + \Sigma_l \delta_l F_{lit-1} + \kappa_i + u_i$ for each of the three sectors separately. B_{ki} is purchases from dependent manufacturing suppliers as a fraction of *Cost of Goods Sold* of the customer firms. F_{li} refers to control variables in the regressions. Please see Appendix B for exact definitions of these variables. In the model reported here, we break down 5-year changes in *Purchases/COGS* into 5 *year-by-year* changes. The regressions also include firm and year dummies but these dummies and the constant are not reported. In each panel, the sample in regression (1) consists of all firms in Compustat between 1979 and 1997 and that in regression (2) consists of firms in Compustat that are identified as a principal customer at least once during the same period. The control variables are censored at the 1st and 99th percentile values to eliminate influence of outliers. Panel A shows the results for the manufacturing firms in the durable goods industries only (primary SIC codes from 3,400 to 3,990), Panel B replicates the analysis for manufacturing firms in the nondurable goods industries only (SIC codes between 2,000 and 3,400), and Panel C shows the result for nonmanufacturing customer firms (all other sectors except the financial sector). The regression for each column is replicated 500 times by drawing firm clusters, with replacement, from the relevant sample.

	(A) Durable Goods Manufacturing Firms						(B) Nondurable Goods Manufacturing Firms						(C) Nonmanufacturing Firms					
	(1)			(2)			(1)			(2)			(1)			(2)		
	Coeff	[95% Conf. Int.]		Coeff	[95% Conf. Int.]		Coeff	[95% Conf. Int.]		Coeff	[95% Conf. Int.]		Coeff	[95% Conf. Int.]		Coeff	[95% Conf. Int.]	
$\Delta_{t-(t-1)}$ Purchases/COGS	−0.010	−0.091	0.072	−0.009	−0.091	0.073	−0.082	−0.204	0.040	−0.087	−0.217	0.043	−0.017	−0.105	0.071	−0.009	−0.107	0.088
$\Delta_{t-(t-2)}$ Purchases/COGS	−0.065	−0.143	0.012	−0.055	−0.144	0.033	−0.166	−0.415	0.083	−0.143	−0.373	0.086	0.038	−0.101	0.176	0.039	−0.106	0.183
$\Delta_{t-(t-3)}$ Purchases/COGS	−0.108	−0.192	−0.025	−0.088	−0.146	−0.031	−0.014	−0.210	0.181	−0.017	−0.210	0.177	0.043	−0.107	0.193	0.039	−0.118	0.196
$\Delta_{t-(t-4)}$ Purchases/COGS	−0.091	−0.192	0.010	−0.077	−0.169	0.014	0.077	−0.118	0.273	0.087	−0.101	0.276	0.119	−0.014	0.252	0.122	−0.018	0.261
$\Delta_{t-(t-5)}$ Purchases/COGS	−0.074	−0.152	0.004	−0.058	−0.126	0.011	−0.004	−0.148	0.140	0.018	−0.113	0.149	0.009	−0.124	0.142	0.001	−0.130	0.132
ΔQ	−0.008	−0.009	−0.006	−0.009	−0.013	−0.005	−0.007	−0.008	−0.005	−0.009	−0.015	−0.003	−0.008	−0.009	−0.006	−0.017	−0.023	−0.011
ΔLog(Sales)	0.016	0.012	0.020	0.018	0.006	0.029	0.016	0.011	0.022	0.009	−0.007	0.024	0.014	0.011	0.018	0.014	0.003	0.026
ΔFA/TA	0.133	0.103	0.162	0.146	0.055	0.236	0.092	0.056	0.127	0.070	0.010	0.130	0.096	0.079	0.113	0.007	−0.053	0.066
ΔROA	−0.033	−0.044	−0.022	−0.046	−0.085	−0.007	−0.024	−0.038	−0.010	−0.030	−0.091	0.031	−0.027	−0.037	−0.018	−0.105	−0.162	−0.048
ΔR&D	−0.012	−0.043	0.019	0.009	−0.101	0.119	0.009	−0.011	0.028	−0.018	−0.094	0.059	−0.005	−0.021	0.011	−0.106	−0.258	0.045
ΔSGA	−0.016	−0.033	0.002	−0.063	−0.121	−0.006	−0.012	−0.023	0.000	0.001	−0.020	0.023	−0.019	−0.033	−0.004	−0.032	−0.088	0.023
ΔDividend	−0.034	−0.041	−0.027	−0.042	−0.060	−0.025	−0.031	−0.041	−0.020	−0.021	−0.049	0.007	−0.027	−0.034	−0.020	−0.044	−0.066	−0.022
ΔS/A	−0.021	−0.027	−0.016	−0.032	−0.050	−0.015	−0.027	−0.034	−0.019	−0.038	−0.063	−0.014	−0.019	−0.024	−0.015	−0.018	−0.029	−0.007
$LDM_{t-5} - LDM^*_{t-5}$	−1.078	−1.111	−1.045	−1.025	−1.085	−0.965	−1.061	−1.103	−1.019	−1.016	−1.098	−0.934	−1.071	−1.096	−1.045	−1.010	−1.085	−0.935
Firms		1,809			324			1,387			221			3,256			300	
Observations		12,452			2,750			9,668			2,064			20,064			2,601	
R-squared		0.69			0.72			0.69			0.68			0.70			0.68	

Table XIA: Year-by-year changes in purchases and five-year leverage changes.

Table XIB

The Effects of Year-by-Year Changes in Sales to Principal Customers on Five-Year Changes in Market Leverage of SupplierFirms

The following model is estimated: $\Delta LDM_{it} = \alpha + \Sigma_k \delta_k \Delta B_{kit} + \Sigma_l \delta_l \Delta F_{lit} + \gamma(LDM_{it-5} - LDM^*_{it-5}) + \kappa_i + u_{it}$, where the market leverage, LDM_{it}, is defined as the ratio of long-term debt to the market value of equity plus total debt. The symbol Δ stands for 5-year changes in the variable. $LDB_{t-5} - LDB^*_{t-5}$ is the deviation of market leverage from its predicted value at year $(t-5)$. LDM^*_{it} is predicted using the model, $LDM_{it} = \alpha + \Sigma_k \delta_k B_{kit} + \Sigma_l \delta_l F_{lit-1} + \kappa_i + u_{it}$ for each of the three sectors separately. B_{ki} is the proportion of sales to each category of principal customer as a percentage of total sales. F_{li} refers to control variables in the regressions. Please see Appendix B for exact definitions of these variables. In the model reported here, we break down 5-year changes, $\Delta BIGALL$, into five year-by-year changes. The regressions also include firm and year dummies but these dummies and the constant are not reported. In each panel, the sample in regression (1) consists of all firms in Compustat between 1979 and 1997 and that in regression (2) consists of firms in Compustat that have been a supplier to a large customer at least once during the same period. The control variables are censored at the 1st and 99th percentile values to eliminate influence of outliers. Panel A shows the results for the manufacturing firms in the durable goods industries only (primary SIC codes from 3,400 to 3,990) and Panel B replicates the analysis for manufacturing firms in the nondurable goods industries only (SIC codes between 2,000 and 3,400). The regression for each column is replicated 500 times by drawing firm clusters, with replacement, from the relevant sample.

	(A) Durable Goods Manufacturing Firms						(B) Nondurable Goods Manufacturing Firms					
	(1)			(2)			(1)			(2)		
	Coeff	[95% Conf. Int.]		Coeff	[95% Conf. Int.]		Coeff	[95% Conf. Int.]		Coeff	[95% Conf. Int.]	
$\Delta_{t-(t-1)}$BIGALL	−0.022	−0.041	−0.004	−0.023	−0.041	−0.006	−0.010	−0.029	0.009	−0.012	−0.030	0.007
$\Delta_{t-(t-2)}$BIGALL	−0.027	−0.049	−0.006	−0.028	−0.049	−0.007	−0.010	−0.033	0.014	−0.013	−0.037	0.012
$\Delta_{t-(t-3)}$BIGALL	−0.021	−0.041	−0.002	−0.022	−0.042	−0.002	−0.001	−0.026	0.024	−0.003	−0.027	0.022
$\Delta_{t-(t-4)}$BIGALL	−0.021	−0.038	−0.003	−0.021	−0.038	−0.004	−0.009	−0.037	0.019	−0.012	−0.040	0.016
$\Delta_{t-(t-5)}$BIGALL	−0.033	−0.049	−0.017	−0.033	−0.049	−0.018	−0.013	−0.036	0.011	−0.014	−0.039	0.012
ΔQ	−0.008	−0.010	−0.007	−0.008	−0.010	−0.006	−0.007	−0.009	−0.006	−0.005	−0.007	−0.003
ΔLog(Sales)	0.017	0.011	0.023	0.014	0.006	0.023	0.018	0.012	0.024	0.017	0.009	0.024
ΔFA/TA	0.147	0.116	0.179	0.162	0.125	0.200	0.098	0.062	0.134	0.120	0.075	0.165
ΔROA	−0.038	−0.051	−0.025	−0.039	−0.054	−0.023	−0.030	−0.045	−0.014	−0.013	−0.032	0.005
ΔR&D	−0.025	−0.062	0.011	−0.043	−0.086	0.000	0.000	−0.022	0.022	0.003	−0.022	0.028
ΔSGA	−0.019	−0.040	0.002	−0.014	−0.041	0.013	−0.017	−0.029	−0.005	−0.012	−0.024	0.001
ΔDividend	−0.035	−0.042	−0.028	−0.031	−0.040	−0.022	−0.028	−0.038	−0.017	−0.033	−0.046	−0.019
ΔS/A	−0.026	−0.032	−0.020	−0.022	−0.030	−0.015	−0.034	−0.042	−0.025	−0.029	−0.038	−0.019
$LDM_{t-5} - LDM^*_{t-5}$	−1.071	−1.104	−1.037	−1.069	−1.110	−1.029	−1.059	−1.102	−1.017	−1.080	−1.133	−1.027
Firms		1,747			1,204			1,332			719	
Observations		12,052			8,167			9,403			4,865	
R-squared		0.70			0.70			0.70			0.69	

Table XIB: Year-by-year changes in big-customer sales and five-year leverage changes.

7.6 Tables XII and XIII: Zero debt and dynamic-panel robustness

Read Table XII:

- The dependent variable is a zero-long-term-debt indicator.
- Durable-goods suppliers with more sales to nongovernment principal customers are more likely to have zero debt.
- The effect is not present for government customers and is weak outside durable goods.

Read Table XIII:

- The Arellano–Bond dynamic panel estimates address concerns about lagged dependent variables with fixed effects.
- The main customer and supplier results remain concentrated in durable-goods sectors.
- The dynamic-panel estimates therefore reinforce the baseline findings.

Economic message: Customer dependence helps explain extremely conservative capital structures, including zero debt, and the main results are robust to dynamic panel estimation.

Table XII

Logit Models of the Effect of Principal Customers on the Probability of the Supplier Firm Having Zero Debt

The following Logit model is estimated: $ZERODEBT_{it} = \alpha + \Sigma_k \theta_{lk} B_{kit} + \Sigma_l \eta_l F_{lit} (+\kappa_i) + u_{it}$, where $ZERODEBT_{it}$ is a dummy variable that takes the value of one if the firm has no long-term debt in that particular year. B_{kit} is the proportion of sales to each category of principal customer as a percentage of total sales. The exact definitions of these variables are in Appendix A. F_{lit} refers to the control variables in the regressions. It includes log of total sales (*Log*(*Sales*)), profitability (*ROA*), tangibility (*FA*/*TA*), growth opportunities (*Q*), R&D to sales (*R&D*), selling general and administrative expense (*SGA*), dividend dummy (*Dividend*), and total sales to total assets (*S*/*A*). The definitions of these variables are in Appendix B. The regressions also include year dummies and firm fixed effects (in column 2s), but their coefficients are not reported. The sample contains all manufacturing firms in Compustat between 1979 and 1997. The control variables are censored at the 1st and 99th percentile values to eliminate influence of outliers. Panel A shows the results for manufacturing firms in durable goods industries only (primary SIC codes from 3,400 to 3,990) and Panel B shows the results for manufactured firms in nondurable goods industries (SIC codes between 2,000 and 3,400). In each panel, results for two sets of tests are reported—those that allow for firm-level clustering (column 1) and those that control for firm-fixed effects (column 2). The *z*-values are given in parentheses. *** (**,*) denotes significance at the 1% (5%, 10%) level.

	(A) Durable Goods Firms		(B) Nondurable Goods Firms	
	(1)	(2)	(1)	(2)
BIGNONGOV	0.388	0.362	0.140	−0.108
	(3.20)***	(2.50)**	(0.95)	(−0.59)
BIGGOVT	−0.240	−0.137	−0.679	−1.082
	(−0.68)	(−0.39)	(−1.17)	(−1.39)
Q	−0.012	−0.032	−0.000	−0.002
	(−1.88)*	(−4.29)***	(−0.04)	(−0.21)
Log(Sales)	−0.361	−0.441	−0.470	−0.498
	(−16.58)***	(−12.32)***	(−16.96)***	(−11.06)***
FA/TA	−4.761	−5.381	−3.571	−5.062
	(−14.75)***	(−17.76)***	(−11.08)***	(−14.17)***
ROA	0.460	−0.138	0.368	−0.300
	(5.07)***	(−1.49)	(3.19)***	(−2.48)**
R&D	0.327	−0.260	0.041	−0.526
	(1.97)**	(−1.26)	(0.25)	(−2.21)**
SGA	0.412	−0.049	0.554	−0.104
	(2.89)***	(−0.31)	(3.65)***	(−0.61)
Dividend	0.590	0.631	0.960	0.334
	(5.39)***	(5.61)***	(8.04)***	(2.38)**
S/A	0.162	0.270	0.229	0.426
	(3.32)***	(5.70)***	(4.94)***	(6.45)***
Constant	−0.680		−0.724	
	(−4.52)***		(−4.05)***	
Firms		1,319		682
Observations	33,295	12,969	25,453	6,814

Table XII: Principal customers and the probability that a supplier has zero long-term debt.

Table XIII

Arellano-Bond Dynamic Panel Data Estimation of the Market Leverage Model

The table reports the results of applying the Arellano-Bond method to our basic model: $LDM_{it} = \alpha + \gamma LDM_{it-1} + \Sigma_k \delta_k B_{kit} + \Sigma_l \delta_l F_{lit} + \kappa_i + u_{it}$, where LDM_{it}, the market leverage, is defined as the ratio of long-term debt to the market value of equity plus total debt. B_{kit} represents purchases from dependent suppliers or sales to large customers. Please see Appendix A for exact definitions of these variables. F_{lit} refers to control variables in the regressions. The definitions of these variables are in Appendix B. The symbol Δ stands for 1-year change in the variable. The regression includes year dummies, but their coefficients are not reported. The sample contains all firms in Compustat between 1979 and 1997. The control variables are censored at the 1st and 99th percentile values to eliminate influence of outliers. Panel A shows the results for the manufacturing firms in durable goods industries only (primary SIC codes from 3,400 to 3,990). In column 2 (respectively, column 3), we drop all customer firms whose size is above median (respectively, below median) among all customer firms. Panel B replicates the analysis for manufacturing firms in the nondurable goods industries only (SIC codes between 2,000 and 3,400) and Panel C shows the result for nonmanufacturing customer firms (all other sectors except the financial sector). The *t*-values, given in parentheses, are based on heteroskedasticity-consistent standard errors. *** (**,*) denotes significance at the 1% (5%, 10%) level.

	(A) Durable Goods Manufacturing Firms				(B) Nondurable Goods Manufacturing Firms		(C) Nonmanufacturing Customers
	Customer Firms Only	Smaller Customer Firms	Larger Customer Firms	Supplier Firms Only	Customer Firms Only	Supplier Firms Only	Customer Firms Only
$LDM_{t-1} - LDM_{t-2}$	0.497	0.433	0.483	0.546	0.549	0.561	0.549
	(11.1)***	(8.07)***	(7.50)***	(19.07)***	(9.81)***	(13.9)***	(10.27)***
ΔPurchases/COGS	−0.042	−0.078	0.073		0.103		0.113
Durable suppliers	(−1.43)	(−2.60)***	(0.70)		(0.72)		(1.73)*
ΔPurchases/COGS	0.138	0.150	−0.138		−0.023		0.024
Nondurable suppliers	(1.10)	(1.13)	(−0.30)		(−0.21)		(0.29)
ΔBIGNONGOV				−0.021		−0.007	
				(−3.21)***		(−0.89)	
ΔBIGGOVT				0.019		0.004	

Table XIII, part 1: Arellano–Bond dynamic panel estimates.

ΔQ	−0.015	−0.010	−0.030	−0.011	−0.013	−0.009	−0.028
	(−6.91)***	(−4.77)***	(−4.70)***	(−13.27)***	(−4.25)***	(−8.80)***	(−5.17)***
ΔLog(Sales)	0.028	0.037	0.043	0.014	0.046	0.019	0.036
	(3.62)***	(4.17)***	(2.83)***	(4.05)***	(4.68)***	(6.04)***	(2.70)***
ΔFA/TA	0.137	0.269	−0.095	0.232	−0.002	0.149	0.098
	(2.29)**	(3.92)***	(−1.20)	(9.57)***	(−0.04)	(5.81)***	(1.45)
ΔROA	−0.049	−0.061	−0.048	−0.028	−0.037	−0.014	−0.099
	(−2.59)***	(−3.03)***	(−1.03)	(−3.61)***	(−1.38)	(−1.56)	(−2.88)***
ΔR&D	−0.018	−0.034	−0.018	−0.008	−0.013	0.007	−0.099
	(−0.33)	(−0.64)	(−0.06)	(−0.56)	(−0.35)	(0.63)	(−1.02)
ΔSGA	−0.051	−0.042	−0.054	−0.027	0.023	−0.015	−0.060
	(−1.76)*	(−1.22)	(−0.84)	(−2.91)***	(0.97)	(−1.95)*	(−1.17)
ΔDividend	−0.010	−0.020	0.006	−0.006	0.023	0.012	−0.009
	(−1.05)	(−1.65)*	(0.51)	(−0.92)	(1.16)	(1.40)	(−0.57)
ΔS/A	−0.047	−0.065	−0.032	−0.039	−0.128	−0.053	−0.054
	(−3.02)***	(−3.53)***	(−1.85)*	(−7.40)***	(−6.78)***	(−7.28)***	(−5.48)***
Constant	−0.006	−0.010	−0.003	0.002	−0.007	−0.013	−0.007
	(−2.00)**	(−1.99)**	(−1.67)*	(0.61)	(−2.80)***	(−6.00)***	(−1.47)

Table XIII, part 2: Continued Arellano–Bond estimates.

Table XIV

Univariate Comparisons of Firm Characteristics after and during Big-Customer Relationships

We compare average values of firm characteristics during the length of a principal customer relationship that lasts for at least 2 consecutive years (labeled "During average"), with the 3-year average immediately after the end of the relationship (labeled "Post average"). Means are reported in the upper row, and medians in the lower row. We perform a paired *t*-test of the means (upper row) and the Wilcoxon matched-pairs signed-rank test of the medians (lower row) for the null hypothesis that post-average minus during-average = 0. In Part I, we report these tests for suppliers of nongovernment big customers in the durable goods sectors and nondurable goods sectors separately. In Part II, we report the results for suppliers of government big customers. *** (**,*) denotes significance at the 1% (5%, 10%) level.

Part I: Average Supplier Characteristics during and after Nongovernment Big Customer Relationship(s)

	(A) Durable Manufacturing Sector				(B) Nondurable Manufacturing Sector			
Firm Characteristics	Post Average	During Average	*t*-Value (*z*-Value)	Obs.	Post Average	During Average	*t*-Value (*z*-Value)	Obs.
Investment-to-Net	0.414	0.536	−4.461***	317	0.321	0.394	−2.095**	200
Fixed Assets	0.306	0.367	(−4.884***)		0.242	0.241	(−1.717*)	
Total Sales	0.123	0.275	−4.934***	213	0.104	0.178	−1.712*	134
Growth[a]	0.099	0.128	(−4.842***)		0.068	0.071	(−1.136)	
Nonbig-Buyer	0.138	0.310	−3.633***	213	0.118	0.194	−1.589**	131
Sales Growth	0.102	0.132	(−3.101***)		0.074	0.073	(−0.377)	
Selling, Gen. &	0.345	0.340	0.354	345	0.265	0.272	−0.297	217
Admin. Exp.	0.306	0.284	(2.054**)		0.184	0.170	(−3.005***)	
Market-to-Book	1.802	2.079	−4.200***	288	1.978	2.071	−1.123	188
Assets	1.378	1.634	(−4.810***)		1.368	1.430	(−1.857*)	
Market-to-Book	2.274	2.702	−3.414***	288	2.410	2.819	−2.690***	179
Equity	1.701	2.088	(−4.705***)		1.757	2.006	(−2.725***)	
Stock Returns	0.093	0.148	−1.634*	266	0.179	0.102	1.807*	162
	0.052	0.096	(−2.194**)		0.135	0.072	(1.981**)	
Altman's *Z*-score	11.188	13.342	−4.142***	291	9.534	12.319	−3.625***	177
(1993)	9.566	10.501	(−4.755***)		7.786	8.653	(−3.292***)	

(*continued*)

Table XIV, part 1: Supplier characteristics after nongovernment big-customer relationships end.

7.7 Table XIV and appendices: Termination evidence and variable definitions

Read Table XIV:

- After nongovernment principal-customer relationships end, suppliers experience declines in investment, sales growth, market-to-book measures, and Altman's Z-score.
- The declines are generally sharper for durable-goods suppliers than for nondurable-goods suppliers.
- After government-customer relationships end, the same adverse patterns are much weaker or absent.

Read Appendix A, Table AI, and Appendix B:

- Appendix A defines the customer- and supplier-related variables used in the regressions.
- Table AI shows the industry distribution of the principal-customer sample.
- Appendix B defines the firm-level control variables and Compustat data items.

Economic message: The termination evidence supports the idea that losing principal customers is economically damaging for suppliers. The appendices clarify that the key regression variables directly map to the paper's stakeholder theory.

Table XIV—*Continued*

Part II: Average Supplier Characteristics during and after Government Big Customer Relationship(s)

	Manufacturing Sector (Both Durable and Nondurable)			
Firm Characteristics	Post Average	During Average	t-Value (z-Value)	Obs.
Investment-to-total	0.406	0.492	−2.424**	250
Net Fixed Assets	0.305	0.319	(−2.589***)	
Total Sales Growth	0.151	0.191	−1.184	188
	0.104	0.087	(−0.300)	
Nonbig-Buyer	0.183	0.165	0.431	186
Sales Growth	0.107	0.065	(1.355)	
Selling, Gen. &	0.339	0.323	0.714	305
Admin. Exp.	0.259	0.245	(2.509**)	
Market-to-Book	2.268	2.306	−0.440	232
Assets	1.621	1.735	(−0.791)	
Market-to-Book	3.009	3.306	−1.606	222
Equity	2.115	2.372	(−1.732*)	
Stock Returns	0.148	0.128	0.499	190
	0.096	0.061	(0.972)	
Altman's Z-score	12.098	13.252	−1.436	228
(1993)	9.279	9.681	(−1.404)	

[a]For total sales growth and nonbig customer sales growth, our starting period is 1 year after the relation terminates. As a result, the number of observations is smaller.

Table XIV, part 2: Supplier characteristics after government big-customer relationships end.

Appendix A: Customer- and Supplier-Related Variables

Dependent Supplier-related Variables:	
Purchases/COGS All identified suppliers	Total supply from all Compustat-listed manufacturing sector suppliers as a proportion of Cost of Goods Sold of the customer.
Purchases/COGS Durable suppliers	Total supply from all durable manufacturing sector suppliers (SIC from 3400 to 3990) as a proportion of Cost of Goods Sold of the customer.
Purchases/COGS Nondurable suppliers:	Total supply from all nondurable manufacturing sector suppliers (SIC from 2000 to 3400) as a proportion of Cost of Goods Sold of the customer.[a]
Big Customer-Related Variables:	
BIGALL	Proportion of sales to all big customers.[b]
BIGNONGOV	Proportion of sales to all of the firm's big customers identified as Compustat firms, private companies or unidentified. Private companies are identified from Academic Universe and they include companies listed elsewhere but not included in Compustat.
BIGGOVT	Proportion of sales to all big customers identified as government agencies.[c]

[a]Purchases/COGS–Durable suppliers and Purchases/COGS–Nondurable suppliers add up to Purchases/COGS–All identified suppliers.
[b]All the components of sales to various categories of big customers are expressed as proportions of the firm's total (net) sales. A reported customer is considered "big" if it accounts for at least 10% of total (net) sales of the firm.

Appendix A: Customer- and supplier-related variable definitions.

Table AI
Industry-wise Distribution of Sample Firms

Two-Digit Sic Code	Industry Description	Number of Firm-Years That Reported Big Customers (1979–1997)	Number of Firm-Years in Compustat (1979–1997)
2,000	Food and Kindred products	772	3,781
2,100	Tobacco products	19	229
2,200	Textile Mill products	452	1,339
2,300	Apparel & other finished products	604	1,574
2,400	Lumber and Wood Pds. Ex. Furn.	188	1,178
2,500	Furniture and Fixture	282	1,069
2,600	Paper and Allied Products	332	1,848
2,700	Printing Publishing and Allied	379	2,579
2,800	Chemicals and Allied	2,635	8,823
2,900	Petroleum refining and related Inds.	125	1,080
3,000	Rubber & Misc. plastic products	828	2,233
3,100	Leather and Leather products	206	459
3,200	Stone, clay, glass, concrete pds.	276	1,288
3,300	Primary metal industries	696	2,641
3,400	Fabr. Metal ex.machy, Trans. Eq.	974	3,150
3,500	Inds. Comml Machy,, Computer Eq.	3,703	10,350
3,600	Electr. Oth. Elec. Eq. ex. Cmp.	4,977	10,870
3,700	Transportation Equipment	1,396	3,435
3,800	Measr. Instr. Photo Gds. Watches	3,313	9,233
3,900	Misc Manufacturing Industries	628	1,829
Total		22,785	68,988

Table AI: Industry-wise distribution of sample firms.

Appendix B: Other Firm Characteristic Variables

Variable Name	Description	Compustat Annual Data Item Number
LDM_t	Total Long-term Debt/(Total Debt + Market Value of Equity)	Item9/[Item6-Item60 + (Item199 * Item25)]
LDB_t	Total Long-term Debt/(Book Value of Total Assets)	Item9/Item6
Q	(Market Value of Equity + Total Liabilities)/ Book Value of Assets	[(Item199 * Item25) + Item6 - Item216]/(Item181+Item216)
Log(Sales)	Natural logarithm of Total Sales (Net).	Log(Item12)
Total Assets	Total Liabilities and Stockholders' Equity	Item6
FA/TA	[Total Fixed Assets (Net)]/ Total Assets	Item8/Item6
ROA	Return on Assets	Item18/Item6
Margin	(Sales(Net)—Cost of Goods Sold)/ Sales(Net)	(Item12- Item41)/Item12
AGE	Number of years the firm has remained in Compustat	
R&D	(Research and Development Expense)/Total Sales (Net)	Item46/Item12
SGA	(Selling, General and Administrative Expenses)/ Total Sales (Net)	Item189/Item12
Dividend	Dummy that is equal to one if the firm pays a common dividend in the particular year; otherwise, zero	Dummy variable based on Item 21
S/A	[Total Sales(Net)]/(Total Assets)	Item12/Item6
CV(Sales)	Coefficient of variation of Total Sales (Net) based on past 5 years' data	SD(Item12)/mean(Item 12)
Advert	Advertising Expense/ Total Sales (Net)	Item45/Item12
Intang	(Intangibles)/(Total Assets)	Item33/Item6
STDE	Standard deviation of a firm's operating income before depreciation, calculated using all available years between 1979 and 1997	*SD* (Item 13)

Appendix B: Other firm characteristic variable definitions.

8 Short summary

- The paper tests whether buyer–supplier relationships affect capital structure.
- The theoretical foundation is the stakeholder theory of capital structure: leverage affects stakeholders' willingness to make relationship-specific investments.
- The empirical setting is U.S. Compustat firms from 1979 to 1997, using principal-customer disclosures.
- The customer-side test asks whether customers that buy more from dependent suppliers choose lower leverage.
- The supplier-side test asks whether suppliers that sell more to principal customers choose lower leverage.
- Results are concentrated in durable-goods industries, where products and assets are more likely to be specific.
- Durable-goods customers have lower leverage when they rely more on durable-goods dependent suppliers.
- Durable-goods suppliers have lower leverage when they sell more to nongovernment principal customers.
- Government principal customers do not have the same effect on supplier leverage.
- Supplier firms dependent on nongovernment customers are more likely to have zero long-term debt.
- When nongovernment customer relationships end, suppliers suffer declines in investment, sales growth, valuation measures, and financial health.
- The paper shows that capital structure can be shaped by product-market stakeholders, not only by taxes, agency costs, and information asymmetry.

9 Conclusion

9.1 Core conclusion

Banerjee, Dasgupta, and Kim show that buyer–supplier relationships matter for capital structure. Firms in durable-goods industries maintain lower leverage when they are involved in relationships where one party depends heavily on the other and relationship-specific investments are likely to be important.

9.2 Economic intuition

Debt increases the probability of financial distress. Financial distress threatens stakeholders who have invested in relationship-specific assets. Because those stakeholders anticipate the risk, firms can preserve relationship value by choosing lower leverage. In this sense, conservative capital structure can serve as a commitment to customers and suppliers.

9.3 Customer-side conclusion

Customers in durable-goods industries use lower leverage when they purchase more from dependent durable-goods suppliers. This is consistent with customers trying to reassure suppliers that they will remain financially healthy enough to support the relationship.

9.4 Supplier-side conclusion

Suppliers in durable-goods industries use lower leverage when a larger share of sales goes to nongovernment principal customers. This is consistent with suppliers protecting themselves against the demand shock and asset redeployability problems that arise if a principal customer relationship ends.

9.5 Incremental contribution revisited

The paper makes the stakeholder theory of capital structure empirically concrete. Instead of treating product uniqueness as an industry-level black box, it uses actual principal-customer links to show that financing policy depends on bilateral economic relationships.

9.6 Implications

The findings imply that optimal leverage depends on supply-chain structure. A firm with many relationship-specific stakeholders may rationally choose less debt than a similar firm with more arm's-length transactions. Capital-structure analysis should therefore consider customer dependence, supplier dependence, asset specificity, and product-market organization.

9.7 Limitations

The evidence is based on observational panel data, and relationship formation is not random. Principal-customer disclosures are incomplete and limited to the pre-1998 disclosure regime. The durable-goods classification is only an imperfect proxy for asset specificity. Still, the pattern of results across durable versus nondurable industries, nongovernment versus government customers, and dynamic robustness checks makes the stakeholder interpretation persuasive.

9.8 Takeaway

The enduring takeaway is that capital structure is not only a financial-market decision. It is also a product-market commitment. Firms use conservative leverage to protect valuable buyer–supplier relationships and to support relationship-specific investment by stakeholders.